7. Applies mu-law of compression
8. Finds and corrects offset
9. Rewrites wave
10. Finds the size of compressed signal

Daubenches wavelet transform performs the following functions:

1. Selects audio and finds the actual signal size
2. Finds amplitude and frequency
3. Chooses a block size
4. Changes compression percentages
5. Initialises compressed matrix
6. Does compression using inverse discrete cosine transform (IDCT)
7. Rewrites signal
8. Finds the size of compressed signal

For complete algorithms, refer code implementations.

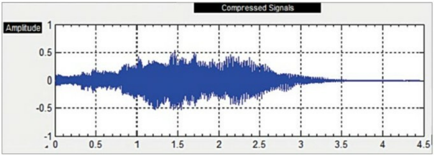

Fig. 4: Daubenches-wavelet-decomposed audio signal (size: 192.043kB)

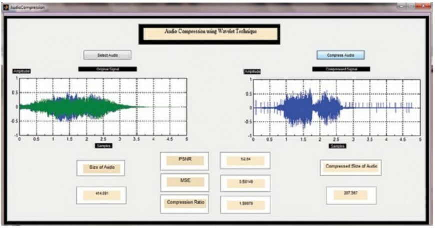

Fig. 5: Graphical user interface for audio compression

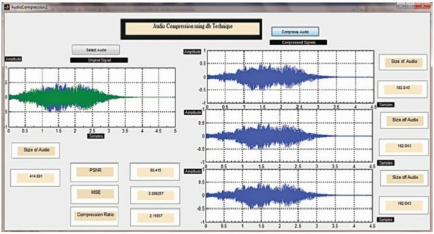

Fig. 6: Program output (Daubenches wavelet)

EFY Note
The source code and equations of this project are included in this month's EFY DVD and are also available for free download at *source.efymag.com*

MATLAB code file AudioCompression.m implements Haar wavelet and AudioCompression2.m file implements Daubenches wavelet. In this example, Windows XP Startup.wav is the sample audio file used for compression.

Comparison of performance metrics such as PSNR, MSE and compression ratio shows that Daubenches algorithm is best suited for lossless compression of speech signals. Advantages of audio compression are less storage space and associated cost, and faster data transfer.

There are several techniques for data compression. You should follow lossless compression technique as lossy audio compression results in data loss.

Image and video can be compressed in a similar way.

Steps to run the program

Run AudioCompresssion.m file. You will get a window as shown in Fig. 1. Select audio file and then press Compress Audio button. You will get output windows as shown in Fig. 5. Compare audio file sizes before and after compression. The compressed audio file is generated as Output1.wav in the same path as the original source file (AudioCompresssion.m).

Now run AudioCompression2.m file. Select audio file and press Compress Audio button. You will get the program output window as shown in Fig. 6. Observe the size of compressed audio. Here, three compressed audio files are generated and saved in the same path as the original source file. These three outputs correspond to different discrete cosine transform (DCT) window sizes of 2, 4 and 8. The percentage change in compressed files may differ depending on the quality of original file and size. EFY

Lalit G. Patil is a software engineer at Bombardier Transportation India. His interests include signal processing and power electronics